How Bright? Apparent and Absolute Magnitudes

What you will be able to do

1. Explain the **magnitude scale** and why it runs "backwards".
2. Convert a **magnitude difference into a brightness ratio**, and back — with or without logarithms.
3. Distinguish **apparent magnitude** m from **absolute magnitude** M.
4. Calculate M from m and distance, and distance from m and M (the **distance modulus**).
5. Compare the true power (luminosity) of stars such as the Sun, Sirius, Rigel and Deneb.
6. Estimate a star's brightness by eye, as variable-star observers do.

1. A 2,000-year-old scale that we still use

Around 130 BCE, **Hipparchus** (Lecture 2.3) sorted the naked-eye stars into **six classes**: **1st magnitude** for the brightest, **6th magnitude** for the faintest he could see. So **bigger numbers mean FAINTER** — it feels backwards, but the scale is too useful to change.

Pogson makes it precise (1856) — an Indian connection

The eye responds to brightness *ratios*: each magnitude step looks like the same *multiple* of brightness. In 1856 the English astronomer **Norman Pogson** fixed the definition used today:

$$\boxed{5 \text{ magnitudes} = \text{a brightness ratio of exactly } 100} \quad \Longrightarrow \quad 1 \text{ magnitude} = \sqrt[5]{100} = 10^{0.4} \approx 2.512.$$

Indian connection — Pogson at Madras

Pogson spent most of his career (1861–1891) as **Government Astronomer at the Madras Observatory** (Chennai). From there he discovered several asteroids, including **87 Sylvia**, and catalogued thousands of stars. His daughter **Isis Pogson** worked with him and was among the first women proposed for fellowship of the Royal Astronomical Society.

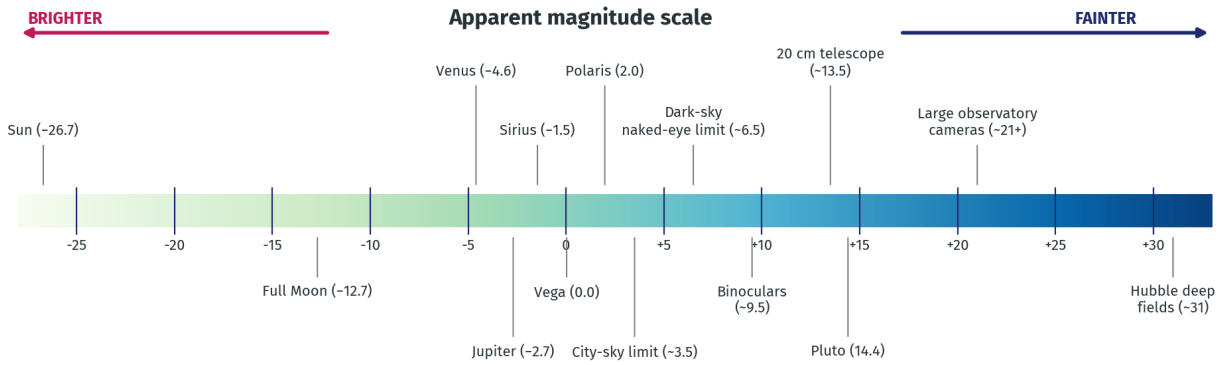
With this definition the scale extends both ways: very bright objects get **negative** magnitudes, and telescopic objects go to +10, +20, +30…

2. Apparent magnitude m: how bright it LOOKS

Apparent magnitude depends on two things: how much light the object really gives out, and how far away it is.

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Every step of 1 magnitude = 2.512× in brightness; 5 magnitudes = exactly 100×. The scale runs BACKWARDS: smaller (or negative) number = brighter.

The apparent magnitude scale, from the Sun to the faintest galaxies photographed by the Hubble Space Telescope.

Object	m	Object	m
Sun	−26.7	Polaris	1.98
Full Moon	−12.7	faintest stars from a big Indian city	~ 3–4
Venus (brightest)	−4.6	faintest naked-eye stars, dark site	~ 6.5
Jupiter (opposition)	−2.5 to −2.9	limit of 10 × 50 binoculars	~ 9.5
Sirius	−1.46	Pluto	~ 14.4
Vega (the old "zero point")	0.03	faintest Hubble galaxies	~ 31

3. From magnitudes to brightness ratios

For two objects with magnitudes m_1, m_2 and brightnesses (fluxes) F_1, F_2:

$$\frac{F_1}{F_2} = 10^{0.4\,(m_2 - m_1)} = 2.512^{\,(m_2 - m_1)}$$

$$m_1 - m_2 = -2.5 \log_{10} \frac{F_1}{F_2}$$

No logs? Use this table

Δm	0.5	1	1.5	2	2.5	3	4	**5**	6	7	10	15
ratio	1.6	**2.5**	4.0	6.3	10	16	40	**100**	251	631	10^4	10^6

Trick: break the difference into fives and ones — every 5 magnitudes multiplies by 100, every 1 by ~2.5. E.g. $7 = 5 + 2 \Rightarrow 100 \times 6.3 \approx 630$.

Worked examples

1. Sirius vs Polaris. $\Delta m = 1.98 - (-1.46) = 3.44$; $\dfrac{F_{\text{Sir}}}{F_{\text{Pol}}} = 10^{0.4 \times 3.44} = 10^{1.376} \approx \mathbf{24}$.

2. Sun vs Full Moon. $\Delta m = -12.74 - (-26.74) = 14$; ratio $= 10^{0.4 \times 14} = 10^{5.6} \approx \mathbf{4 \times 10^5}$ — four lakh times!

3. A star 100 times fainter than Vega ($m = 0$) is 5 magnitudes fainter: $m = \mathbf{+5}$.

Worked examples *(continued)*

4. One star is 4 times as bright as another: $\Delta m = 2.5 \log_{10} 4 = 2.5 \times 0.602 = \mathbf{1.5}$ magnitudes.

The inverse-square law

Move a light source twice as far away and its light spreads over 4 times the area:

$$F \propto \frac{1}{d^2} \qquad \Longrightarrow \qquad 2\times \text{ farther} = 4\times \text{ fainter} = 2.5\log_{10} 4 \approx 1.5 \text{ magnitudes fainter.}$$

4. Absolute magnitude M: how bright it REALLY is

To compare stars fairly, imagine placing every star at the **same standard distance of 10 parsecs**:

> **Absolute magnitude** M = the apparent magnitude a star would have **at a distance of 10 parsecs**.

Units of distance. 1 parsec (pc) is the distance at which a star shows a parallax of $1''$:

$$1 \text{ pc} = 3.26 \text{ light-years} = 206{,}265 \text{ AU} \approx 3.09 \times 10^{13} \text{ km}, \qquad 10 \text{ pc} = 32.6 \text{ ly}.$$

The distance modulus

$$\boxed{M = m - 5\log_{10}\left(\frac{d}{10 \text{ pc}}\right)} \quad \Longleftrightarrow \quad m - M = 5\log_{10} d - 5 \quad \Longleftrightarrow \quad \boxed{d = 10^{(m-M+5)/5} \text{ pc}}$$

$m - M$ is the **distance modulus**: every 5 magnitudes of it means **10 times farther**.

Worked examples

5. The Sun. $m = -26.74$, $d = 1$ AU $= 1/206{,}265$ pc:

$$M = -26.74 - 5\log_{10}\left(\frac{1}{2{,}062{,}650}\right) = -26.74 + 31.57 = +\mathbf{4.83}.$$

Moved to 10 pc, the Sun would be a faint 5th-magnitude star — barely visible from a city!

6. Sirius. $m = -1.46$, $d = 2.64$ pc: $M = -1.46 - 5\log_{10}(0.264) = -1.46 + 2.89 = +\mathbf{1.43}$.

7. Rigel. $m = 0.13$, $d \approx 264$ pc: $M = 0.13 - 5\log_{10}(26.4) = 0.13 - 7.11 = -\mathbf{7.0}$ — a supergiant.

8. Distance from magnitudes. A Sun-like star ($M = 4.83$) appears at $m = 9.83$: $d = 10^{(9.83-4.83+5)/5} = 10^2 = \mathbf{100}$ **pc** (326 ly).

How much more powerful than the Sun?

$$\boxed{\frac{L}{L_\odot} = 10^{0.4\,(4.83-M)}}$$

Star	m	Distance	M	$L/L_\odot$ (visual)
Sun	−26.74	8.3 light-min	+4.83	1
Sirius	−1.46	8.6 ly	+1.43	∼23
Vega	0.03	25 ly	+0.60	∼50
Arcturus	−0.05	37 ly	−0.3	∼110
Polaris	1.98	∼430 ly	−3.6	∼2,400
Betelgeuse	∼ 0.5	∼550 ly	∼ −5.6	∼14,000
Rigel	0.13	∼860 ly	**−7.0**	∼ 53,000
Deneb	1.25	∼2,600 ly	**−8.3**	∼ 170,000

The big lesson: Deneb looks fainter than Sirius but is ∼7,000 times more luminous — it is ∼300 times farther away. Brightness in the sky tells you little until you know the distance.

5. Colour and magnitude

Astronomers also measure magnitudes through coloured filters, e.g. **B** (blue) and **V** (visual). The **colour index** $B - V$ measures colour: ≈ 0 for white stars (Vega); negative for hot blue stars (Spica ≈ -0.23); large and positive for cool red stars (Betelgeuse $\approx +1.85$) — matching the star colours you observed in Unit 2.

6. Things that change magnitudes

- **Atmospheric extinction:** near the horizon, starlight passes through more air and dust; in hazy Indian skies a star at 10° altitude can look 1–2 magnitudes fainter.
- **Light pollution** brightens the sky background, hiding faint stars.
- **Variable stars** really change: **Algol** dims from 2.1 to 3.4 every 2.87 days as a companion eclipses it; **Mira** ranges from ∼3 to ∼10 over about 11 months; **Betelgeuse** faded to ∼1.6 in its "Great Dimming" of 2019–20.

Estimating a star's magnitude by eye (Argelander's step method)

Pick a comparison star A slightly brighter and B slightly fainter than the target V. Judge, e.g., "V is 2 steps fainter than A and 1 step brighter than B". Then

$$m_V = m_A + \frac{2}{2+1}\,(m_B - m_A).$$

Amateurs worldwide send such estimates to the **AAVSO** (American Association of Variable Star Observers) — real science anyone can do.

Key points to remember

- Smaller (or negative) magnitude = brighter. 5 mag = 100×; 1 mag = 2.512×.
- $F_1/F_2 = 10^{0.4(m_2-m_1)}$; $m_1 - m_2 = -2.5\log_{10}(F_1/F_2)$.
- Pogson defined the scale (1856) and later directed the Madras Observatory.
- M = brightness at 10 pc: $M = m - 5\log_{10}(d/10)$; $d = 10^{(m-M+5)/5}$ pc.
- Sun: $m = -26.74$, $M = +4.83$. 1 pc = 3.26 ly. Doubling the distance adds 1.5 magnitudes.

Check yourself

1. How many times brighter is a magnitude 1 star than a magnitude 6 star?
2. A star has $m = 3$ and $M = 3$. How far away is it?
3. Which looks brighter, $m = -0.5$ or $m = +0.5$? By what factor?
4. If the Sun were moved to 100 pc, what would its apparent magnitude be?

Answers: 1. 100×. 2. 10 pc. 3. $m = -0.5$, by $10^{0.4} \approx 2.5$×. 4. $m = 4.83 + 5\log_{10} 100 - 5 = 9.83$ — binoculars needed.

Further reading (free / public)

- OpenStax, *Astronomy 2e*, §17.1 "The Brightness of Stars" and §19.1 "Fundamental Units of Distance" — `https://openstax.org/books/astronomy-2e/pages/17-1-the-brightness-of-stars`
- AAVSO, *Visual Variable Star Observing Manual* (free) — `https://www.aavso.org`